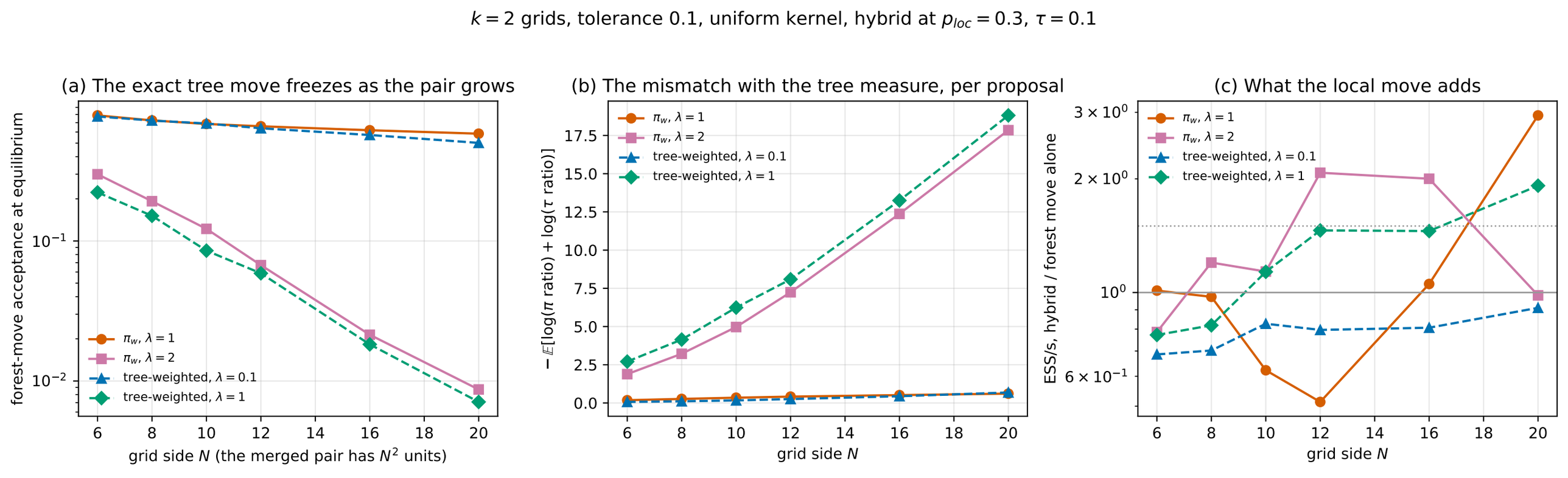
$k = 2$ grids, tolerance 0.1, uniform kernel, hybrid at $p_{loc} = 0.3$, $\tau = 0.1$
(a) The exact tree move freezes as the pair grows
forest-move acceptance at equilibrium
10^{-1}
10^{-2}
π_w, $\lambda = 1$
π_w, $\lambda = 2$
tree-weighted, $\lambda = 0.1$
tree-weighted, $\lambda = 1$
6 8 10 12 14 16 18 20
grid side N (the merged pair has N^2 units)
(b) The mismatch with the tree measure, per proposal
$-\mathbb{E}[\log(\pi \text{ ratio}) + \log(\tau \text{ ratio})]$
17.5 15.0 12.5 10.0 7.5 5.0 2.5 0.0
π_w, $\lambda = 1$
π_w, $\lambda = 2$
tree-weighted, $\lambda = 0.1$
tree-weighted, $\lambda = 1$
6 8 10 12 14 16 18 20
grid side N
(c) What the local move adds
ESS/s, hybrid / forest move alone
3×10^0
2×10^0
10^0
6×10^{-1}
π_w, $\lambda = 1$
π_w, $\lambda = 2$
tree-weighted, $\lambda = 0.1$
tree-weighted, $\lambda = 1$
6 8 10 12 14 16 18 20
grid side N